

Lecture 7: Impulse, Momentum and Collisions – Supplementary Notes

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These notes support Lecture 7. They contain derivations and extensions that were moved out of the contact time.

1 The impulse–momentum theorem for a varying force

For a net force that varies with time, $\sum \vec{F} = d\vec{p}/dt$ integrates directly: $\int_{t_1}^{t_2} \sum \vec{F} dt = \vec{p}(t_2) - \vec{p}(t_1)$. Hence the impulse is the area under the $F-t$ graph, and the average force over $\Delta t = t_2 - t_1$ is $\vec{F}_{av} = \vec{J}/\Delta t$.

2 The elastic collision formulas

A mass m_1 moving at v_1 collides elastically and head on with m_2 at rest. Momentum: $m_1 v_1 = m_1 v'_1 + m_2 v'_2$. Kinetic energy: $\frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_1 v'^2_1 + \frac{1}{2} m_2 v'^2_2$. Rearranging,

$$m_1(v_1 - v'_1) = m_2 v'_2, \quad m_1(v_1^2 - v'^2_1) = m_2 v'^2_2.$$

Dividing the second by the first (for $v'_2 \neq 0$): $v_1 + v'_1 = v'_2$, which says that the relative velocity reverses, $v_1 - 0 = -(v'_1 - v'_2)$. Substituting $v'_2 = v_1 + v'_1$ in the momentum equation gives

$$v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1, \quad v'_2 = \frac{2m_1}{m_1 + m_2} v_1.$$

3 Kinetic energy lost in a completely inelastic collision

For two objects that stick together, the final velocity is $v_{cm} = (m_1 v_1 + m_2 v_2)/(m_1 + m_2)$, and the kinetic energy lost is

$$\Delta K = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (v_1 - v_2)^2,$$

which depends only on the *relative* speed.

4 The centre of mass

For particles with masses m_i at positions $\vec{r}_i$, $\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{M}$ with $M = \sum m_i$. Differentiating, $\vec{P} = M \vec{v}_{cm}$; differentiating again and using Newton's third law to cancel internal forces gives $\sum \vec{F}_{ext} = M \vec{a}_{cm}$. In a collision of an isolated system, $\vec{v}_{cm}$ is the same before and after. The centre of mass returns in the lecture on rotation.

5 Rocket propulsion

A rocket expels exhaust at speed v_{ex} relative to itself. In a short time it ejects a small mass; conservation of momentum gives $M dv = -v_{ex} dM$ (with $dM < 0$), so, with no external force,

$$v - v_0 = v_{ex} \ln \frac{M_0}{M}.$$

The final speed grows with the exhaust speed and the *logarithm* of the mass ratio, which is why rockets are mostly fuel.